Marco Antonio Adame Rodríguez

IoT & Embedded Systems Engineer · Digital Signal Processing · Scientific Software

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Profile

I build end-to-end IoT systems: MEMS sensor, embedded firmware, real-time acquisition, signal processing, Linux backend, cloud and dashboards. I designed a low-cost structural health monitoring system from scratch and validated it in the field against commercial seismic instrumentation (GEA and P-Alert); the work led to three peer-reviewed papers and an international conference talk. I also build production geospatial web applications (Supabase, Leaflet, offline-first) with automated tests and continuous deployment. Computer Engineer and MSc graduate in Engineering for Innovation and Technological Development (UAGro), 9.42/10 GPA, degree conferral expected November 2026.

Key achievements

- **Complete SHM-IoT system designed from the ground up** and validated in the field against commercial GEA and P-Alert equipment, at a fraction of the cost of traditional seismic instrumentation.
- **Ambient vibration measurement on a biaxial shaking table:** synchronized 200 Hz sampling, FFT analysis over 2048-point windows, 0.98 Hz frequency resolution, and automatic RPM computation from the dominant frequency (peer-reviewed publication).
- **Continuously running multi-site monitoring:** nodes across separate networks, a centralized Ubuntu Server receiver, local buffering against connectivity loss, and published web dashboards.
- **Python/PyQt5 vibration analysis suite** that processes large volumes of records and generates comparative technical reports automatically.
- **Production seismic-risk web platform:** real-street routing in 5 Baja California cities to 370 health facilities from the official CLUES registry, offline support, and 152 automated tests on every change.

Research & development experience

Autonomous University of Guerrero (UAGro) · MIIDT graduate program, Lead Researcher and Developer — SHM-IoT System — Aug 2024 – Present

End-to-end design and implementation of a low-cost structural health monitoring solution for civil works, field-validated against commercial GEA and P-Alert equipment.

- Designed acquisition nodes on Arduino Nano RP2040 Connect and Raspberry Pi 4, integrating four MEMS sensors (MPU9250, MPU6050, LSM6DSOX and ADXL335) over I2C, SPI and UART.
- Implemented synchronized 200 Hz acquisition across both RP2040 cores, with ample Nyquist margin for the structural frequency range of interest.
- Validated the system in five environments — a biaxial shaking table, university buildings, a botanical garden, a high school campus and lightweight experimental structures — comparing its records against commercial GEA and P-Alert equipment.
- Built a Python (NumPy/SciPy) DSP pipeline to identify the dynamic response of structures: RMS, FFT, power spectral density (Welch), dominant frequency, coherence, phase analysis and cross-correlation.
- Developed PyQt5 desktop applications for acquisition, live monitoring and analysis, with session control and automatic crash recovery.
- Built tools to process large TXT, CSV, JSON and JSONL record sets: time synchronization, resampling, gap detection, digital filtering and series trimming.
- Automated the generation of technical reports and comparative plots against reference equipment; these results underpin my lead-author paper.
- Designed a centralized multi-site receiver over HTTP/HTTPS with local buffering, so connectivity outages do not cause data loss.
- Developed REST APIs in FastAPI for ingesting, persisting and querying ESP32 node telemetry.
- Administered continuously running Ubuntu servers: systemd services, Docker containers, SSH/VNC remote access, backups and synchronization.
- Published web dashboards on Vercel to visualize, filter and download data from both custom nodes and commercial sensors in one place.

Technical skills

Languages & frameworks: Python, C/C++ (embedded), JavaScript, Node.js, SQL, PHP, Bash, PyQt5 / PySide6, React, Preact

Web & mapping: Leaflet, GeoJSON, OpenStreetMap, PWA / Service Workers, Browser testing (Puppeteer)

Data & signals: NumPy, Pandas, SciPy, Matplotlib, FFT, PSD/Welch, RMS, Digital filters, Time series

AI & machine learning: Machine Learning, Deep Learning, Computer Vision, Big Data, Feature selection

IoT & hardware: Raspberry Pi, ESP32, Arduino Nano RP2040, MEMS sensors, I2C, SPI, UART

Backend & cloud: FastAPI, REST APIs, Supabase, PostgreSQL, Firebase, Vercel Functions

Systems & DevOps: Linux / Ubuntu Server, Docker, systemd, SSH, Git, GitHub Actions (CI), Vercel

Projects

SHM-IoT MIIDT — Low-cost structural monitoring
Full architecture for acquisition, storage, analysis and visualization of accelerations in civil structures, field-tested against commercial sensors.
- Python · C++ · Raspberry Pi · PyQt5

Tijuana seismic risk ecosystem
Web dashboard linking seismic hazard, soil, stations, acceleration maps, inspection and outreach on a single Supabase database. Computes routes to hospitals and emergency services over real streets (OpenStreetMap, Dijkstra) in five Baja California cities using the official CLUES health registry, works offline as an installable app, and ships with semantic versioning and automated tests on every change.
- JavaScript · Leaflet · Supabase · PWA · GitHub Actions

Green Infrastructure · Baja California
Dashboard of 36 green-infrastructure techniques per municipality, with point queries over GeoJSON layers and address search, a login-protected editing panel, PDF and CSV reports, offline mode and a Supabase visit counter.
- JavaScript · Preact · Supabase · Vercel Functions · GeoJSON

Geo-referenced SHM monitoring network
Web dashboard showing location and status of distributed monitoring nodes, continuously deployed and public.
- JavaScript · Vercel

Supabase sensor dashboard
Web application for submitting, querying and visualizing measurements stored in the cloud.
- JavaScript · Supabase · PostgreSQL

ESP32 telemetry APIs
REST services for receiving, persisting and querying acceleration data, plus scientific plotting tools.
- Python · FastAPI

Publications & talks

Ambient Vibration Measurement System on a Biaxial Vibrating Table (Lead author)
Adame Rodríguez, M. A., Alcaraz Vázquez, M., Martínez, E. T., García Guevara, D. S., Alonso Silverio, G. A.
revistas.udistrital.edu.co/index.php/vinculos/article/view/24610 (Revista Vínculos, 22(1), 2026)

Feature-importance-based attribute selection for dengue case classification in Mexico (Co-author)
Alcaraz Vázquez, M., **Adame Rodríguez, M. A.**, García Guevara, D. S., Martínez, E. T., Alonso Silverio, G. A.
revistas.udistrital.edu.co/index.php/vinculos/article/view/24567 (Revista Vínculos, 22(2), 2025)

IoT-based biological sample shaker with temperature control (Co-author)
García Guevara, D. S., Martínez, E. T., Alcaraz Vázquez, M., **Adame Rodríguez, M. A.**, Alonso Silverio, G. A.
revistas.udistrital.edu.co/index.php/vinculos/article/view/24607 (Revista Vínculos, 22(2), 2025)

Conference talk

Conference talk: "Ambient Vibration Measurement System on a Biaxial Vibrating Table". 15th International Computing Conference (CICOM 2025), Tunja, Colombia, October 2025.

Education

Universidad Autónoma de Guerrero, MSc in Engineering for Innovation and Technological Development — Aug 2024 – 2026
- GPA 9.42 / 10 · All coursework completed · No failed courses
- Terminal specialization: Information and Communication Technologies (2023 curriculum). All coursework completed; degree conferral in progress, expected November 2026. Thesis: "Development and implementation of a low-cost IoT system for structural health monitoring in civil works".
- **Artificial intelligence and data:** Machine Learning (10), Deep Learning (10), Computer Vision (10), Data Analysis and Big Data (10)
- **Embedded systems and IoT:** Selected Topics in Embedded Systems (10), IoT Systems Design (9), Fundamentals of Electronic Systems (9)
- **Programme core:** Information and Communication Technologies (10), Sustainable Innovation and Technological Development (10), Graduate Research I (8), Graduate Research

II (8), Graduate Research III (9), Professional Residency (completed), Graduation Project (completed)

Universidad Autónoma de Guerrero, BSc in Computer Engineering 2018 – 2023

- GPA 8.56 / 10 · 55 courses passed
- School of Engineering, Chilpancingo. Degree awarded 2023. Federal professional license no. 13426767.
- **Programming and software engineering:** Programming Fundamentals (9), Object-Oriented Programming I (10), Object-Oriented Programming II (8), Advanced Programming (8), Data Structures I (10), Data Structures II (9), Computational Logic (8), Systems Analysis and Design (10), Software Engineering (8), Translators and Interpreters (8), Compilers (9), Mobile Application Development (10), Human-Computer Interaction (9)
- **Hardware, electronics and architecture:** Digital Systems (10), Computer Organization (10), Server Architecture (10), Microcontrollers (9), Microprocessors (8), Electrical Circuits (9), Electricity and Magnetism (10), Electronics (7)
- **Networking, operating systems and security:** Routing Fundamentals (10), Operating Systems I (10), Operating Systems II (8), Network Fundamentals (9), Structured Cabling Systems (9), IT Resource Auditing (9), Network Security (8), Computer Forensics (8), Communications Fundamentals (7)
- **Data, artificial intelligence and signals:** Databases II (9), Databases I (7), Artificial Intelligence Fundamentals (7), Digital Image Processing (7)
- **Mathematics and basic sciences:** Differential and Integral Calculus (8), Probability and Statistics (8), Numerical Methods (8), Analytic Geometry (8), Operations Research (8), Basic Chemistry (8), Vector Calculus (7), Differential Equations (7), Algebra (7), General Physics (7)
- **Research and general education:** Research Seminar I (9), Research Seminar II (8), Entrepreneurship Workshop (10), Managing Information and Communication Technologies (10), Logical, Heuristic and Creative Thinking (10), Communication Skills (10), Contemporary World Analysis (10), English I (7), English II (7)
- **Practicum:** Professional Internship (completed), Social Service (completed)

Additional experience

Ministry of Labour (Mexico), Jóvenes Construyendo el Futuro — Publishing department Jun 2023 – Jun 2024
12-month vocational training programme in digital production and printing.

35th Military Zone (Mexican Army), Professional internship — IT area Aug 2022 – Jan 2023
450 certified hours. Computer equipment support and maintenance in an institutional environment.

Ignacio Manuel Altamirano Secondary School, Social service — Computing area Feb 2022 – Jul 2022
500 certified hours. Technical support and community-sector educational support.

Certifications

- Data Science Essentials with Python — Cisco / UAGro (Dec 2025)
- Cybersecurity Essentials — Cisco (Mar 2022)
- NDG Linux Unhatched — Cisco (May 2022)
- CCNAv7: Introduction to Networks — Cisco (Jun 2021)

Languages

- Spanish: native.
- English: intermediate (B1), TOEFL ITP 470 (2025). Fluent reading of technical documentation and scientific literature.